Notes: Goldberg, Khandelwal, Pavcnik, and Topalova (AER, 2009)

Trade Liberalization and New Imported Inputs

Contents

1 Big picture

- **Question.** Did India's post-1991 trade liberalization mainly expand imports along the intensive margin, or did it give firms access to genuinely new imported inputs and varieties? How large are the implied gains from these newly available inputs?
- **Setting.** The paper studies India's import data from 1987 to 2000, spanning the period before and after the 1991 trade reforms. The analysis uses product-level customs data by source country and focuses especially on imported inputs for manufacturing firms.
- **Core challenge.** A rise in imports after liberalization could just mean that India bought more of the same goods from the same countries. The important question is whether reform instead relaxed technological constraints by making previously unavailable products and varieties accessible.
- **Main conceptual contribution.** The paper separates import growth into three pieces:
  - new *products* (new HS6 codes),
  - new *varieties* within existing products (new country–HS6 pairs),
  - and growth in already imported country–product pairs.

  In a developing-country setting, the product margin can matter a lot, not just the variety margin.
- **Main hypotheses.**
  - Trade reform should expand access to imported inputs through the extensive margin.
  - These new inputs should be especially important for capital, intermediate, and basic goods rather than only final consumption goods.
  - If reform relaxed technological constraints, many of the new inputs should come from advanced countries and look higher quality than existing imports.
  - New varieties should reduce the effective imported-input price index once variety gains are taken into account.
- **Main empirical design.** The paper uses Indian import values and quantities by product and partner country, aggregates mostly to the HS6 level, classifies goods by end use, decomposes import growth between 1987 and 2000, and then computes Feenstra-style variety indices for machinery categories.
- **Main baseline result.** Overall Indian import growth is dominated by the extensive margin. New products and new varieties account for about **82%** of total import growth between 1987 and 2000, and for imported inputs the corresponding share is about **86%**.
- **Key heterogeneity result.** The extensive-margin expansion is strongest for imported inputs—basic, capital, and intermediate goods—rather than for final goods. For many input categories, the product-extensive margin alone explains most of the growth.

- **Key source-country result.** A large share of new imported products comes from OECD countries. That pattern is strongest for input categories, suggesting that liberalization opened access to technologically more advanced suppliers.
- **Key price result.** In companion calculations emphasized here, variety growth deflates India's conventional imported-input price index by about **38%** between 1989 and 1997, or about **4.7% per year**.
- **Main credibility point.** The paper uses a conservative HS6 classification, the original 1987 coding to avoid false product turnover, and a distinction between products and varieties that is especially important for a developing country such as India.
- **Economic takeaway.** India's trade reform did not merely make existing imports cheaper or larger in volume. It appears to have expanded the menu of imported production inputs, especially from advanced countries, thereby lowering effective input prices and plausibly easing domestic firms' product-creation constraints.

2 Data and measurement (institutional context and key objects)

2.1 Why India and the 1991 reform are the right laboratory

- The paper is motivated by India's broad trade liberalization during the 1990s after a long period of import substitution and tight restrictions.
- In related work by the same authors, two-thirds of the surge in imported inputs came from products not imported before the reforms.
- Over the same decade, about one-quarter of Indian manufacturing output growth was associated with new domestic products.

Interpretation. India is useful because the trade reform was large and because the pre-reform regime plausibly kept many inputs and product varieties out of the country.

2.2 Import data and sample construction

- The data come from official Indian import records from Tips Software Services.
- The underlying records report import values and quantities by source country at the eight-digit Harmonized Tariff System (HS) level from **1987 to 2000**.
- The analysis is conducted mainly at the **HS6** level, which contains about 5,000 product codes and is standardized internationally.
- The authors deliberately use the **original 1987 HS classification** so that measured product turnover does not simply reflect later revisions of product codes.

Interpretation. Moving to HS6 makes the exercise more conservative. It reduces the chance that India-specific coding detail exaggerates the amount of variety growth.

2.3 Products versus varieties

The paper distinguishes sharply between a *product* and a *variety*:

- A **product** is an HS6 category.
- A **variety** is an HS6–country pair.

For example, HS6 854220 (hybrid integrated circuits) is a different product from HS6 854280 (electronic integrated circuits not elsewhere specified), while a hybrid integrated circuit imported from Japan is a different variety from the same HS6 product imported from Germany.

Interpretation. In rich-country data, most action is often on the variety margin because those countries already import most products. In India's setting, whole product categories were previously absent, so the product margin matters too.

2.4 Imported inputs versus final goods

The paper classifies each HS6 code into end-use categories following Nouroz (2001), which relies on India's input-output matrix. The main grouping is:

- **final goods:** consumer durables and consumer nondurables;
- **inputs:** intermediate goods, capital goods, and basic products.

Interpretation. This classification lets the paper ask whether liberalization mainly increased imports of goods used in production rather than only consumer goods.

2.5 Main growth objects

For a product group g, let real import growth between 1987 and 2000 be

$$G_g = 100 \times \frac{M_{g,2000} - M_{g,1987}}{M_{g,1987}}, \tag{1}$$

where imports are deflated by wholesale price indices.

The paper decomposes this growth into three components:

$$G_g = G_g^{\text{prod-ext}} + G_g^{\text{var-ext}} + G_g^{\text{int}}, \tag{2}$$

where:

- $G_g^{\text{prod-ext}}$ is growth due to **new HS6 products** imported in 2000 but not in 1987;
- $G_g^{\text{var-ext}}$ is growth due to **new country–product pairs** within HS6 products already imported in 1987;

- G_g^{int} is growth in **existing country–product pairs**.

Interpretation. The paper's central descriptive question is whether G_g is mostly an extensive-margin phenomenon.

2.6 OECD versus non-OECD source countries

The product- and variety-extensive margins are further split by source country:

$$G_g^{\text{prod-ext}} = G_{g,\text{OECD}}^{\text{prod-ext}} + G_{g,\text{non-OECD}}^{\text{prod-ext}}, \tag{3}$$

$$G_g^{\text{var-ext}} = G_{g,\text{OECD}}^{\text{var-ext}} + G_{g,\text{non-OECD}}^{\text{var-ext}}. \tag{4}$$

Interpretation. If advanced-country suppliers account for most newly imported inputs, that supports the idea that trade reform relaxed technological constraints rather than merely expanding low-end sourcing.

2.7 Variety-adjusted price index

For a sector s, the paper uses the Feenstra (1994) and Broda-Weinstein (2006) methodology to compute a variety index:

$$\Lambda_s = \left(\frac{\lambda_{s,1997}}{\lambda_{s,1989}}\right)^{\frac{1}{\sigma_s - 1}}, \tag{5}$$

where $\lambda_{s,t}$ is the expenditure share on varieties common to both periods and σ_s is the elasticity of substitution across varieties.

Interpretation.

- A lower Λ_s means larger gains from new imported varieties.
- If new varieties account for a bigger share of expenditure in 1997 than in 1989, the conventional price index overstates the true price of imported inputs.

2.8 Descriptive facts

- India imported **3,249** HS6 products and **23,571** varieties in 1987.
- By 2000 these counts had grown to **4,443** products and **55,819** varieties.
- The average number of source countries per product increased from **7.3** to **12.6**.
- Overall import growth between 1987 and 2000 was **130 percentage points**.
- Inputs grew much faster than final goods: **227** versus **90** percentage points.
- For machinery (HS84), the mean variety index across HS4 codes is **0.861**, the median is **0.911**, the minimum is **0.196**, and the maximum is **1.135**.

Interpretation. The raw data already indicate a major widening in India's import menu, both across product categories and across supplying countries.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual mechanism and hypotheses

The paper's mechanism comes from trade and growth models in which foreign inputs raise productivity in two ways:

- **static gains:** firms gain access to cheaper or previously unavailable inputs;
- **dynamic gains:** access to new inputs lowers the cost of innovation and helps firms create new domestic products.

Interpretation. The paper itself is mainly about the import-side objects—new products, new varieties, and price-index gains—that make these broader productivity channels plausible.

3.2 Accounting decomposition of import growth

For any broad product group g, the empirical decomposition is:

$$G_g = G_g^{\text{prod-ext}} + G_g^{\text{var-ext}} + G_g^{\text{int}}. \tag{6}$$

The meaning of each term is:

- **product extensive margin:** imports of HS6 codes not imported at all in 1987;
- **variety extensive margin:** imports from new source countries within HS6 products already imported in 1987;
- **intensive margin:** additional imports in country–product pairs already active in 1987.

Interpretation. This is not a regression model. It is a decomposition exercise. The key empirical object is the share of import growth that comes from newly available inputs.

3.3 Why separate products from varieties

The literature often emphasizes varieties, where a variety is a product imported from a particular country. But in a developing-country setting a product itself may be entirely absent before reform. That motivates:

$$\text{Extensive margin} = \text{new products} + \text{new varieties within existing products.} \tag{7}$$

Interpretation. If one only measured country-level variety growth, one would miss an important portion of liberalization's effect in India.

3.4 OECD decomposition

The paper further decomposes the extensive margin by country origin:

$$G_g^{\text{prod-ext}} = G_{g,\text{OECD}}^{\text{prod-ext}} + G_{g,\text{non-OECD}}^{\text{prod-ext}}, \tag{8}$$

$$G_g^{\text{var-ext}} = G_{g,\text{OECD}}^{\text{var-ext}} + G_{g,\text{non-OECD}}^{\text{var-ext}}. \tag{9}$$

Interpretation. This split asks whether the newly available input menu was disproportionately supplied by richer countries, which would be consistent with access to more advanced or differentiated technologies.

3.5 Variety-adjusted import price index

The Feenstra-Broda-Weinstein methodology adjusts the conventional import price index by the contribution of new varieties. In reduced form, the price index for sector s is multiplied by the variety term

$$\Lambda_s = \left(\frac{\lambda_{s,1997}}{\lambda_{s,1989}}\right)^{\frac{1}{\sigma_s - 1}}. \tag{10}$$

Interpretation.

- $\lambda_{s,t}$ is smaller when expenditure shifts toward varieties not present in the base period.
- If $\Lambda_s < 1$, new varieties lower the true price index relative to the conventional one.
- The article emphasizes that, for imported inputs overall, this adjustment implies about a **38%** decline from 1989 to 1997, or **4.7% annually**.

3.6 Machinery as the case study sector

The paper zooms in on HS84 (nuclear reactors, boilers, and machinery) because it likely contains important capital inputs used by many manufacturing industries. The variety index is computed at the HS4 level within HS84.

Interpretation. The machinery sector provides a concrete illustration that the newly imported varieties were not just statistically present; they were economically meaningful in a key class of production inputs.

3.7 What the article does not estimate

- The paper does *not* estimate a tariff elasticity of import variety in this short article.

- It does *not* run firm-level production regressions here.
- It does *not* separately identify the causal effect of imported input access on domestic product creation within this paper.

Instead, it documents the composition of import growth and then interprets those patterns in light of the authors' companion papers.

Interpretation. The paper's strength is careful descriptive measurement of the extensive margin, not a stand-alone causal production equation.

4 Identification and empirical design (what makes the evidence informative)

4.1 Why the paper is descriptive but still valuable

- The article is mainly an accounting and measurement exercise around the reform period.
- Its purpose is to show *how* imports changed after liberalization, not just *how much*.
- The evidence is informative because the decomposition directly targets the margin that trade-and-growth models care about: access to previously unavailable inputs.

4.2 Why the HS6 focus is conservative

- HS6 codes are internationally standardized.
- Working at HS6 reduces the chance that measured product turnover reflects India-specific coding detail.
- The authors explicitly note that this choice likely biases measured variety growth downward.

Interpretation. If the extensive margin looks large even at HS6, the true growth in available varieties is unlikely to be smaller.

4.3 Why the original 1987 classification matters

- Product codes are sometimes revised over time.
- If later classification changes were treated as new products, measured product turnover would be mechanically inflated.
- By anchoring the analysis to the original 1987 HS classification, the paper isolates "true" turnover more cleanly.

Interpretation. This is one of the paper's most important design decisions.

4.4 Why product versus variety distinction is crucial in a developing country

- In rich countries, extensive-margin trade growth usually comes from new source-country varieties within products already imported.
- In India, many products themselves were not imported before reform.

Interpretation. The paper's distinction between products and varieties is not cosmetic. It is central to understanding how liberalization changed a previously constrained economy.

4.5 Why the end-use classification matters

- The classification into final goods, basic products, capital goods, and intermediate goods comes from an input-output based end-use mapping.
- This avoids arbitrary ex post labeling of "important" inputs.

Interpretation. When the paper finds the largest extensive-margin growth in imported inputs, that pattern is tied to a systematic economic classification.

4.6 Why source-country composition strengthens the interpretation

- Most of the product-extensive growth for inputs comes from OECD countries.
- New OECD and non-OECD varieties also tend to have higher unit values than existing varieties within the same HS6 categories.

Interpretation. These facts are consistent with new imported inputs being more technologically advanced or more differentiated than the old ones.

4.7 Why the machinery variety evidence matters

- The machinery sector plausibly contains capital inputs used widely across industries.
- Large variety gains in this sector are therefore economically meaningful even without a direct firm-level production regression in the article.

Interpretation. The case study makes the broader decomposition feel concrete: liberalization changed the effective menu of productive equipment available to firms.

4.8 What the paper can and cannot establish

- It *can* show that import growth was dominated by newly available products and varieties, especially for inputs.

- It *can* show that new inputs often came from advanced countries and generated nontrivial variety-adjusted price declines.
- It *cannot* by itself prove the magnitude of firm-level productivity gains within this short article.
- It *cannot* fully separate tariff reform from all other contemporaneous changes in India's economy.

Interpretation. The article is strongest as a measured description of the import-side channel through which trade reform may generate static and dynamic gains.

4.9 Relation to the literature

The paper sits at the intersection of three literatures:

- trade liberalization and productivity,
- gains from imported product varieties,
- and endogenous-growth models in which foreign inputs facilitate innovation.

Its distinctive contribution is to show that, in a developing country, the *product margin* and the *input margin* of imports are empirically central.

5 Tables and figures

5.1 Table 1: overall decomposition of import growth

Read:

- India's total imports rose by about **130 percentage points** between 1987 and 2000.
- Of that increase, about **84** points come from **new HS6 products**.
- Another **22** points come from **new country varieties** within existing HS6 products.
- Only about **23** points come from growth in already existing country–product pairs.

Economic message. Most of India's import expansion was not "more of the same." It was driven by products and varieties that were simply not present before.

5.2 Table 1: final goods versus inputs

Read:

- Final goods imports grew by about **90** percentage points; inputs grew by about **227**.
- For final goods, the decomposition is roughly **33** from new products, **25** from new varieties, and **32** from the intensive margin.

TABLE 1—EXTENSIVE AND INTENSIVE MARGIN OF INDIA'S IMPORTS, 1987–2000

	Import growth	Product extensive margin			Variety extensive margin			Intensive margin
		Total	OECD	Non-OECD	Total	OECD	Non-OECD	Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All products	130	84	59	25	22	9	13	23
Final products (consumer durables and nondurables)	90	33	21	11	25	9	16	32
Inputs (capital, basic, intermediates)	227	153	115	38	42	15	26	32
Basic products	260	154	124	30	62	31	31	45
Capital products	125	37	27	10	33	23	10	55
Intermediate products	297	278	200	78	28	−12	39	−9
HS Code 27 (mineral fuels and oil)	89	59	0	59	11	1	11	19
HS Code 28 (inorganic chemicals)	227	7	4	2	92	4	88	128
HS Code 29 (organic chemicals)	158	4	2	2	58	12	46	95
HS Code 71 (precious stones and metals)	668	666	576	89	28	20	8	−25
HS Code 72 (iron and steel)	27	34	16	18	24	4	20	−31
HS Code 84 (nuclear reactors, boilers, and machinery)	100	33	23	10	27	21	6	39
HS Code 85 (electrical machinery and equipment)	173	72	63	9	35	19	16	66

Notes: The table decomposes import growth into the extensive and intensive margins between 1987 and 2000. The first column reports overall import growth. Column 2 reports the contribution to import growth due to the extensive (new HS6) margin. Columns 3 and 4 disaggregate column 2 according to the source country. Column 5 reports the contribution to growth due to existing HS6 codes. This product extensive margin is decomposed into the variety extensive margin (column 5) and the variety intensive margin (column 8). Columns 2, 5, and 8 sum to column 1. The variety extensive margin is decomposed in the variety extensive margins in columns 6 and 7. All variables are deflated by wholesale price indices. Please see footnote for the list of OECD countries. The units in the table are percentage growth in import values (rupees).

- For inputs, the decomposition is roughly **153** from new products, **42** from new varieties, and **32** from the intensive margin.

Economic message. Liberalization changed the production side of the economy most strongly. Imported inputs, not just final goods, expanded through the extensive margin.

5.3 Table 1: basic, capital, and intermediate goods

Read:

- Basic products grew by about **260** percentage points, with **154** from new products and **62** from new varieties.
- Capital goods grew by about **125** points, with **37** from new products and **33** from new varieties.
- Intermediate goods grew by about **297** points, with **278** from new products and **28** from new varieties.

Economic message. Every major input class shows large extensive-margin growth, and for intermediates the product margin is extraordinarily important.

5.4 Table 1: OECD versus non-OECD origin

Read:

- Overall, OECD countries account for about **59** of the **84** product-extensive growth points in total imports.
- For basic, capital, and intermediate goods, OECD countries supply more than **70%** of product-extensive growth.
- Within existing HS6 products, OECD countries account for about **41%** of new variety growth overall.

Table 2—Variety Index within Nuclear Reactors, Boilers, and Machinery (HS 84)

HS4 Code	Variety index
8471 Automatic data process machines	0.196
8479 Machines having individual functions	0.940
8473 Parts for office machines	0.319
8443 Printing machinery	0.822
8445 Machines for preparing textile fibers	0.940
8406 Steam turbines	0.899
8416 Furnace burners, mechanical stokers, etc.	0.978
8428 Lifting, handling, loading, and unloading machinery	0.744
8429 Self-propelled bulldozers, graders, shovels, etc.	0.774
8438 Machinery for industrial preparation of food and drink	0.747
8439 Machinery for making pulp	0.834
8453 Machinery for working leather	0.903
8462 Machine tools for forging, bending, etc.	0.702
8475 Machines for assembling electric tubes, etc.	0.816
8454 Converters, ladles, and casting machines	0.873
Minimum (8471 automatic data process machines)	0.196
Maximum (8476 vending machines)	1.135
Median variety index	0.911
Mean variety index	0.861

Notes: This table reports the variety index developed by Feenstra (1994) for selected HS4 codes within sector HS 84 between 1989 and 1997. The top panel reports the five largest HS4 codes in 1997. Summary statistics computed over the 85 possible HS4 codes within HS 84 are reported in the bottom panel. Estimates for the elasticity of substitution are from Broda, Greenfield, and Weinstein (2006), who estimate India's elasticities of substitution at the HS3 level.

- The paper also reports that new OECD varieties are about **2.7%** more expensive than existing OECD varieties, while new non-OECD varieties are about **5.5%** more expensive than their existing counterparts.

Economic message. Newly available inputs were not random. They disproportionately came from more advanced countries and look more differentiated or higher quality.

5.5 Table 1: detailed sectors

Read:

- The bottom panel of Table 1 looks at fuels, chemicals, precious stones and metals, iron and steel, machinery, and electrical machinery.
- Even in these narrower sectors, new products and new varieties account for a very large share of import growth.
- For HS84 machinery, for example, total growth is about **100**, with **33** from new products and **27** from new varieties.
- In some sectors the intensive margin is small or even negative, meaning the expansion is dominated by newly available goods rather than old trade relationships.

Economic message. The extensive-margin story is not an artifact of broad aggregation. It survives in narrow, economically important sectors.

5.6 Table 2: variety index within machinery (HS84)

Read:

- The lowest variety index in HS84 is for **automatic data processing machines (HS8471)**, with an index of **0.196**.
- Machine tools for forging and bending metal (HS8462) have an index of about **0.702**.
- Across HS84 categories, the **mean** variety index is **0.861** and the **median** is **0.911**.
- The paper emphasizes that a lower variety index means larger gains from new varieties.

Economic message. New imported machinery varieties materially reduced the effective price of capital inputs, especially in some key equipment categories.

5.7 Price-index implication from the variety calculations

Read:

- In the broader companion calculations summarized in the article, variety growth lowers India's conventional overall import price index by about **31%** between 1989 and 1997.
- For *imported inputs* specifically, the deflation is about **38%**, or roughly **4.7% per year**.

Economic message. Ignoring new imported varieties would substantially overstate the true cost of imported inputs faced by Indian firms.

6 Short summary

- The paper studies how India's imports changed after trade liberalization.
- Using product-by-country customs data, it shows that most import growth came from *new products* and *new varieties*, not just more of the same imports.
- This extensive-margin expansion is much stronger for *imported inputs* than for final goods.
- A large share of the new input menu comes from OECD countries, consistent with access to more advanced suppliers.
- New imported varieties also reduce the effective imported-input price index substantially.
- The paper's broader message is that trade reform can benefit domestic firms by expanding the set of productive inputs they can use.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that India's trade liberalization was associated with a dramatic expansion in imported products and varieties, especially for goods used as production inputs. The dominant margin of import growth was extensive, not intensive.

7.2 Economic intuition

When trade barriers fall in a previously protected economy, firms do not just buy larger quantities of old inputs. They gain access to entirely new products and new source-country varieties. That broader menu can lower effective input prices, relax technological bottlenecks, and make new domestic products easier to create.

7.3 Takeaway

The key lesson is that the gains from trade in developing countries can come heavily from the arrival of *new imported inputs*. Measuring only average tariff changes or the value of preexisting imports misses an important part of what liberalization actually changes.